2019 Theory Examiners' Report

a) Theory Chair Report – Neil Tully MW

This year's theory exam produced 33 passes, representing a 32% pass rate (2018 = 31.25%).

This continues the trend of the last few years which indicates that many candidates are well prepared across all five theory papers. A good number of candidates this year passed the theory exam with high marks across all papers; really excellent papers written with authority, clarity and conviction.

Importantly, marks across all five theory papers were generally consistent at all levels. This is in contrast to previous years where the examiners often saw quite uneven performance in the theory exam with, for example strong marks on Papers 1, 2 and 3, and much lower marks on Papers 4 and 5, or conversely, great Papers 4 and 5 and much less successful "technical" papers 1, 2 and 3. This suggests that candidates are better prepared across the whole exam, and this is to be commended.

However, there were a significant number of candidates with marks some way below the pass mark, this was often on account of not only a general lack of appropriate knowledge but also a failure to address the questions being asked. This is a matter raised by examiners on all theory papers.

Candidates are reminded to first read the question very carefully. Much thought and consideration is put into the wording of our questions, candidates should be under no illusion as to what is being asked.

Very often the reason for candidates not achieving better marks is that the question answered was not the same as the question being asked. Candidates must be honest with themselves; do not write an answer to the question you *want* to see, but write an answer to the question that is on the page in front of you. Do not be willfully blind about this. The examiners know exactly what they are looking for, and obfuscation will not work.

Questions will sometimes ask for critical analysis. This is also an aspect that is often avoided or overlooked. This element is often at the heart of the question, it is the analysis that will demonstrate "mastery" to the examiners, therefore candidates who do not address this element of any question will seriously limit their potential marks. The theory examiners on all papers are looking for evidence of *applied* knowledge.

Candidates approaching the exam with English not as their first language are advised to consider very carefully the best option of whether to have papers translated or to write in English.

b) Theory Exam Paper One - Joel Butler MW

General Comments

The Paper 1 examiners were pleased to see a slight uplift in the overall standard of responses to this year's paper; although Question 2 was the least well-answered question, the examiners saw many well thought through answers to the question set.

While the examiners welcome appropriate use of examples, these should be specific and used to demonstrate a point being made in response to the question, rather than being used as "generic" examples that do little to serve the candidate's purpose in answering the question. These tend to indicate an overly "bookish" learning, rather than real world understanding.

Definition of terms needs to be carefully judged. While precision is important, it is not necessary to slavishly define every term in each question.

The examiners urge candidates to check that facts being used are accurate. Information is available from many sources, and all may not necessarily be reliable. The examiners noted a number of inaccurate "facts" from possibly questionable sources.

Q1. **Are yield restrictions necessary to produce high-quality wine?**

A successful response to this question required finding meaningful connections between various methods of yield restriction and their impact on wine quality and by doing so arriving at a conclusive answer. A more "opinion-editorial" approach would have been unlikely to pass.

Generally, the question was well answered; most candidates managed to grasp the essential understanding that it was about vineyards where yield manipulation was consciously required. Correctly, they did not spend much time writing about low yielding vineyards based upon old vines, restrictive soils, etc. Poor time management was an issue for a number of candidates, evidenced by a paucity of writing, shortage of examples, or leaving out discussion of a key issue such as canopy management.

The examiners in particular were looking to see where candidates, in evaluating "are" (yield restrictions necessary) were able to demonstrate "how" (they might be necessary to produce high-quality wine) in order to answer the question successfully.

The best answers also noted clearly that with new viticultural techniques, new clones, better understanding of vine balance and the effect of climate change in certain locales, low yields are not as necessary to producing high-quality wines as they once were. More importantly, there is no 'one size fits all' for lowering yields to produce high-quality grapes.

Q2. Your company has acquired a vineyard which has been producing grapes for bulk wine. How would you convert the vineyard to high-quality wine production?

This question was approached from a number of different angles by candidates, some choosing to focus on a single, notional vineyard and others writing in more general terms. Both could have been successful as long as they considered the main principles and importantly focused on *conversion*. Many answers rolled out good information on how to set up and farm a vineyard for high quality, but didn't necessarily address the real-world issues of converting a vineyard from one purpose to another.

A number of responses that discussed step by step how to establish a vineyard missed the point of the question, which is about converting. Many answers missed chances to show mastery and real-world understanding by too quickly assuming that replanting was the only way to go. It was important to consider the viticulture issues of conversion: changing the canopy (management), manipulating water availability/delivery, altering vine density, pruning, row alignment etc.

Examiners found too many papers where the argument presented for conversion used examples not related to a real bulk wine vineyard type. One paper noted that Chambolle-Musigny producer (hardly bulk wine area) Frederick Mugnier said that "while spur pruning can be used successfully on young vines, it is better in time to convert them to cane-pruning to restrict the yields and achieve more terroir-driven wines." A similar notation from an overachieving grower in the Midi would have been more accurate and justifiable.

Q3. How can drought tolerance be achieved in viticulture?

This was not a question about climate change.

Candidates were expected to evaluate the viticultural techniques available to achieve drought tolerance of vines. It was relevant to consider how learnings from regions that have long operated in near-drought conditions, such as Napa, Mendoza and parts of Australia, may be applied in vineyards experiencing changing climatic conditions, or indeed in new regions that have hitherto been considered unsuitable for viticulture due to existing drought conditions. Being Theory Paper 1, the emphasis should be on viticulture.

Under consideration would be recent research and technologies such as infra-red aerial photography, as well as important viticultural issues such as adaptive rootstocks, cultivar change, cover crop and canopy management, improved irrigation regimes and changing appellation restrictions in non-permitted irrigation areas, not forgetting the issue of actual water availability in areas where irrigation is permitted.

Passing papers provided solid discussion and examples from around the globe for at least five to six of these viticultural issues, presented in a balanced manner. The best papers interlinked various tactics, reflecting the fact that no one single aspect could be solely responsible for managing drought tolerance.

Less successful responses omitted one or more critical issues such as canopy management, soil analyses/amendments, or cover crops, and dwelt strongly on water management or selecting drought-tolerant scions. Very few papers discussed using new hybrids or crosses. The very small percentage of papers that discussed the important research being done at viticultural research centers on the subject indicated a significant knowledge gap.

Q4. Mildews continue to blight vineyards. What strategies might a vineyard manager implement to reduce the risk?

This was a question that could only be answered well if one had a good knowledge of the standard remedies. Examiners were primarily looking for an informed discussion about what are considered the two most important mildews: downy mildew and powdery mildew. Botrytis and Phomopsis could have been included in the discussion, as long as the two primary mildews were more strongly considered.

The best papers noted that there are many fungicides and treatments available and that they must be rotated, or not all used together, in order to ensure that the mildews don't develop resistance. It was important to consider not just copper and sulphur treatments, but also to acknowledge and explain how alternative, resistant varieties, canopy management, vineyard hygiene, climate modelling/forecasting, and the critical timing of treatments can reduce mildew damage.

While no one was expected in this paper to be intimate with the exact formulas, candidates should have considered the differences between contact, local systemic and fully systemic treatments; few addressed the different modes of action that lie behind a diverse, sustainable approach to preventing mildew.

The Examiners were looking for a wide-ranging discussion of different strategies, including organic/biodynamic approaches, in particular noting that there is no easy organic/biodynamic substitute for using copper sulphate, which is why it is accepted (although now in the EU at much reduced rates).

Q5. What are the most critical considerations for selecting rootstock when establishing a new vineyard?

This question was the best answered on the paper. As with Question 4, it required specific discussion and clear examples that demonstrated the writer's knowledge of rootstock traits, limitations and consequences for their use. Examiners were looking for papers that properly addressed matters including:

1. A primary consideration for rootstocks is protection from pests in the soil such as phylloxera and nematodes.

2. Rootstocks are used to address specific challenges or limitations of a site, such as soil chemistry, or excess/inadequate water, or soil structure/fertility.

3. The compatibility and health status of the scion.

4. How vine vigour can be impacted; therefore requiring consideration of variety, spacing, vine size, soil nutrients, salinity.

5. How rootstock choices can affect yield, ripening time, quality and flavour which, in turn, strongly influence the end goal of wine quality.

6. How the relationship between all of these must be taken into consideration: there is no simple, single answer to choosing a rootstock.

Good papers had specific examples that supported their discussion, rather than simply enumerating facts about rootstocks without context. Many papers, for example, noted that nematode presence was a critical consideration for selecting a rootstock, offering some examples. But then didn't analyse why, and for what kind of nematode and potential disease impact they may vector, and under what circumstances.

Good papers also delineated interconnected reasons for using a rootstock; for example to deal with nematodes and drought tolerance. Several papers also considered how climate change may require new rootstock use in traditional areas where changing circumstances such as drought conditions require a new approach.

On the other hand, a number of non-passing papers provided few to no examples of rootstocks needed for particular issues. Or, if the main reasons for using rootstocks were listed, there was often a lack of analysis addressing the question: the key considerations when planting a new vineyard.

Q6. Discuss the role of the following factors for producing high-quality grapes:

- **Aspect**
- **Vine density**
- **Row orientation**

Generally, questions such as this offer an opportunity to write on a specific issue and show real mastery. It was important to bear in mind, however, that this question asked for the issues to be addressed specifically in the context of high-quality grape production.

Some candidates appeared to have run out of time on this question, possibly as a result of it being the final question on this paper to have been approached. Even with limited time available, to achieve good marks it was essential to cover each of the three elements of the question, as the marks were allocated equally between them.

Some responses simply did not cover sufficient detail, giving just a standard Diploma-type answer with very few examples and very little demonstration of understanding or analysis.

On the positive side, excellent papers clearly noted the intimate relationship of slope and sun exposure in a higher latitude site, such as the Graacher Himmelreich on the Mosel, or considered the issue of wind damage control as a key factor when considering row orientation and density, beyond the obvious issue of sunlight exposure, changing from a N-S to an E-W line.

c) Theory Exam Paper Two - Meg Brodtmann MW

General Comments

Examiners made a number of comments that apply to all questions on this year's Theory Paper 2.

Examples should be used to support a statement rather than as the statement itself. There was a very high occurrence of examples cited from the Napa Valley and Bordeaux; more than one examiner commented on the lack of diversity in the use of examples. Candidates should have an understanding of what happens in some of the less obvious regions.

Brain dumps and winemaking lists were also commented on – as is the case every single year. Students must demonstrate critical thinking if they are to pass. Students must also understand the costs related to winemaking procedures. Typos were and are a problem – please reread your answers before submitting.

Students are reminded this is a technical paper so dosage rates are important and must be realistic. Simply saying 'a sufficient amount of x should be added' is not enough – students must demonstrate competence in their understanding of additions and their timings.

Definitions in the introduction are vital and set the scope of the essay. Please do not use abbreviations unless you have previously defined it (e.g. reverse osmosis (RO)).

Some obvious examples of time-keeping issues appeared. It is still very important to plan the essay effectively, be concise, use relevant and interesting examples, answer the question and do not run out of time!

Q1. How can a winemaker manage the impact of oxygen during the winemaking process?

In many cases, candidates did not have the depth of knowledge that was required to answer this question successfully, whilst those that did shone brightly with a number of excellent scripts. Structure in this question was crucial as there was a great opportunity to waffle about oxygen in so many forms. The best answers really focused on the question and were very competent.

The challenge was to get to grips with how winemakers can *manage* oxygen, responses that failed to confront this stood out quickly. There were too many formulaic answers with a simple regurgitation of facts rather than providing an actual answer to the question as set.

Many candidates concentrated on what oxygen does rather than discussing how a winemaker can *manage* the impact of oxygen. We were not just asking the effect of oxygen on wine but the techniques to manage those outcomes. Candidates are urged to cover the entire winemaking process in an orderly manner and cover a range of wine styles. The best candidates used a methodical approach where they discussed the impact and then how that impact could be managed. It is important to remember that oxygen contact has both negative and positive impacts.

Q2. Evaluate the options available to the winemaker wishing to make wine with a lower level of alcohol.

Relatively few candidates attempted this question, but those who did generally provided strong responses. Those that were not as successful tended to put forward a list of the processes involved without sufficient critical analysis on the pros and cons of the various methods – the question clearly stated 'evaluate', not 'tell us all about how to make wine with a lower level of alcohol'.

What did we want? Options that should have been considered included picking dates, site selection, varieties, water additions, yeast choice, vessel type, stopping fermentation, and post fermentation options such as blending, dealcoholisation via spinning cone, reverse osmosis and vacuum distillation.

Please keep up to date with changes to winemaking laws. For example, Australia can now add water to juice/must to lower the potential alcohol to no lower than 13.5%.

Q3. Explain the procedures that might be followed in the winery to deal with rot-degraded fruit.

This was a question which prompted a good number of responses, however candidates did not score well if they were not fully informed on the issues at hand. It was not possible to produce a satisfactory response based only on partial knowledge. The answer needed some careful planning and a few scripts seemed to rush headlong into answering without evidence of a structured approach. A lack of structure sometimes resulted in main points being missed from answers. Examiners also commented on a narrow range of examples only being used.

A number of candidates gave us a paragraph or two on noble rot; this was not required. The best answers covered the whole winemaking process from reception to end of maturation. Vineyard choices were not to be considered as the question clearly stated 'in the winery'.

Candidates should have given a definition of what constitutes rot-degraded fruit. Red and white grapes may have been covered separately. If candidates had broken down the winemaking process in to its parts it would be possible to clearly define what measures could be taken. Thermovinification should have been a consideration.

Q4. With new French oak barrels becoming increasingly expensive, what alternative options and techniques are available to the winemaker wishing to make high-end wine?

A question answered by a good number of candidates, but very often using examples limited to only Bordeaux and the Napa Valley, which does not indicate a truly global perspective.

This was not just an oak alternative question; plenty of candidates gave us great information on different oak sources, different wood types and wood formats, but very few discussed other vessel types (with/without oxygen transfer rates), use of lees, tannin, white wines, sparkling wines, Madeira and different ways of getting oxygen contact with the wine.

Very few candidates managed to define ‘high-end wine’ – concentration, longevity, complexity, balance. These factors are not only seen in red wines.

The best responses considered what a French oak barrel can give to a wine and then discussed alternatives to achieve the same end result. These candidates also discussed techniques to add complexity and concentration to wines that did not involve oak contact. A good response to this question requires more than a list of which winemaker is doing what.

Q5. Consider the impact of rosé winemaking techniques on wine quality. Are paler-coloured rosé wines better quality?

Generally, students struggled with this question with very little consideration being given to the “impact of rosé winemaking techniques on wine quality" part of the question. Most candidates focussed on colour extraction without considering other winemaking processes that impact on wine quality. As a result, many aspects of the specific winemaking processes for rosé were ignored. Picking time and grape varieties were largely ignored as were fermentation, residual sweetness and techniques for adjusting colour. Temperature of time on skins was rarely mentioned, despite it having a big effect on colour extraction.

Lack of definitions in the introduction was often a problem. There was also very little reference to the actual impact of winemaking on quality. As a Paper 2 question, a response from a marketing / Paper 4 point of view would not have been appropriate.

This was a question where candidates should have covered the entire winemaking process in a logical manner, commenting on how various processes impact on quality.

Q6. To what extent is it possible for producers of tank method sparkling wines to match the style and quality of wines produced by the traditional method?

This was a clear question, avoided by many, which exposed a number of candidates’ lack of up-to-date knowledge of sparkling wine production of all types and styles.

Too many candidates did not answer the actual question, preferring to write about how Prosecco and Champagne are made without referring to the question of quality and at which steps quality could be influenced throughout the *entire* production process. Where Sekt was mentioned, most candidates took for granted that Sekt is always made from Riesling which demonstrated a lack of awareness of today’s premium Sekt production with Burgundian varieties.

Quality in sparkling wine production does not just come from ageing on lees; every part of the winemaking process can impact on the final quality of the wine and candidates should have covered this. As a result, very few considered the fact that any given method is not a guarantee of quality – it is the process within those methods that determines the outcomes.

d) Theory Exam Paper Three - Caroline Gilby MW

General Comments

Many of the questions on this year's Theory Paper 3 required thought and analysis rather than simply trotting out a pre-prepared essay. As ever, strong candidates demonstrated appropriate technical knowledge, supported by real world evidence and examples. However, too many lacked factual knowledge or the ability to adapt their answers.

One examiner commented that a number of candidates appeared to reproduce pre-prepared essays vaguely relevant to the question, without focussing sufficiently on the perspective of the question. Such a strategy is unlikely to be successful, as any such answer is likely to include much that is irrelevant and to omit much that is relevant. Another complaint was that some candidates limited the focus and ground to be discussed in their introduction, but without justification.

Examiners also commented that too often examples were not well used; they tended to be 'dropped in' briefly rather than incorporated into a critical discussion. There was also a number of otherwise good candidates who limited their examples to the Southern Hemisphere rather than showing evidence of global wine knowledge.

It bears repeating that Paper 3 is a technical paper and it is expected that candidates will have a firm grasp of some key factual and numerical information. EU limits for SO2 use and units, being clear about whether the numbers refer to total or free SO2; typical levels of free SO2 for various styles; actual values for pH; permitted and actual usage levels for common wine additives; pore sizes for filtration – this is all key knowledge for MWs. It is worth double-checking these too - numbers like 100 g/l SO2 or 45 micron filters cropped up. Vague comments like "lower levels of SO2" are not sufficient – actual numbers are required. Inaccurate use of "sulphur" instead of sulphites or sulphur dioxide still crops up regularly, and while this may be widespread shorthand among winemakers, there are important differences between sulphur as an element and its compounds. Another bugbear is candidates who believe that "brett" is bacterial – *Brettanomyces* is a yeast species and yeast are a completely different kingdom of life to bacteria. Similarly, TCA is not a bacterial problem and VA is not a microbial problem in itself but may be the result of microbial metabolism. Unfortunately, errors like these undermine confidence in the candidate. Technical knowledge and topics such as QA/QC may not be the most glamorous part of the wine industry, but they are critical in ensuring wine reaches the consumer in sound condition.

In general candidates who did not pass wrote superficial answers that were descriptive rather than analytical, lacked examples and numbers. There were also a few frustrating candidates who wrote well but didn't answer the question so could not score good marks.

Q1. Explain which pre-bottling treatments and QC analyses you would consider most important for an inexpensive, organic, vegan wine.

Key understanding:

- *Need for pre-bottling treatments and QC analyses in general; risks if not undertaken – practical risks in terms of impact on wine, and commercial and legal/regulatory risks.*
- *Market positioning and category (inexpensive, vegan, organic); factors affecting choices of treatments, taking account of relevant regulation and cost of treatments.*
- *Awareness of production pressures of inexpensive wine – likely to be high volume, quicker winery throughput (no time for barrels, natural clarification etc).*
- *Awareness of legislation for organic wine – must comply with EU or other applicable rules which specifies permitted additives and processing techniques (e.g. no electrodialysis, reverse osmosis etc). Lower sulphite levels than non-organic wine.*
- *Understanding of implications for vegan claim – no animal products even as processing aids.*

This question required candidates to think about implications of the specified parameters. Most candidates did understand that the question was meant to cover one wine with all these parameters, but some considered three different wines. An inexpensive wine should have been defined. Markets do vary, but a 20 Euro wine is hardly inexpensive for most. It was not helpful to use examples of premium wines such as classed growth Bordeaux or Ridge to illustrate this answer. It was also reasonable to expect that an inexpensive wine needed to be stable, clear, free from hazes and deposits and possibly would contain residual sugar, and would likely need to be prepared quickly and in large volumes.

Many answers were let down by lack of awareness of permitted additives and treatments under organic regulations – so proposing electrodialysis, CMC, sorbic acid, PVPP or Velcorin were errors. Knowledge of permitted sulphite levels, at least for EU and other key markets like USA, was required. Once again, some candidates used sulphur and sulphite as interchangeable, which is incorrect.

Good candidates discussed the implications of reduced protection at lower SO_2 levels and the need to pay attention to hygiene, filtration and pH management. A number of candidates were not aware of what vegan meant – it's not just about ingredients but also any animal derived products as processing aids even if not detectable in the final wine. This has implications for traceability at all stages and limits the range of fining products that can be used. Candidates writing poor answers lacked factual knowledge and tended to write general QC answers rather than thinking about the question and in some cases, time was wasted covering earlier parts of the winemaking process.

Q2. What steps should a winemaker take, in preparation for bottling and at bottling, to prevent microbial spoilage? Consider both red and white wines.

Key understanding:

- *Microbial spoilage can occur at any stage of the winemaking process.*
- *Prevention is better than cure. In many situations cure is not possible.*
- *Hygiene is the number one rule in preventing microbial spoilage.*

This question required a definition of microbial spoilage. Microbiological spoilage organisms can be any which are unwanted at a particular stage (but typically bacteria and yeast) and may include fermentation yeast or malolactic bacteria that had been desirable during winemaking but can cause spoilage in bottle. This includes those organisms which produce off-flavours, odours, colours, or precipitates, or refermentation in bottle, or have the potential to do so. Such faults could include volatile acidity (VA), ethyl acetate (EA), Brettanomyces, taints, mousiness, geranium taint and yeast spoilage. Cork taint is not considered microbial spoilage. The risks differ for red and white wines primarily because red wines have no/very low levels of malic acid and are usually fermented dry.

It was much harder to write a well-structured essay without defining the major risks and implications and many candidates simply wrote about microbial spoilage in general, which tended to result in superficial answers. Particular weaknesses noted included confusing free, total and molecular SO2; errors on filtering and pore sizes; confining the answer to SO2 management; lack of coverage of hygiene, gas management and temperature control; confusing VA with Acetobacter.

As one examiner highlighted, “It has been raised previously but the same applies again this year – a technical question requires a technical answer.” Here one of the more pertinent points was free and total sulphur dioxide. In very simple terms, free sulphur dioxide levels at bottling should be between 25 – 35mg/L or 0.8ppm molecular. Special cases e.g. bulk wines or dessert wines may go up to 40mg/L. It is of great concern that 56 candidates, or 65% of all that answered this question, either provided levels outside this range, didn’t mention any level or discussed total sulphur dioxide as the preventative level for microbial spoilage.

In addition, lots of essays suggested candidates think winemakers use a lot of chemical processes to make/finish their wines rather than ‘good old fashioned’ attention to detail, sound judgement, and hygiene control or oxygen management. The chemical processes that were regularly mentioned as standard practice include DMDC, sorbic acid, velcorin, lysozyme and chitosan. A considerable number of candidates implied that applying these and other additives at the time of bottling was normal practice rather than showing/explaining remedial steps and processes prior to getting it to this point. Trying to fix, in the case of this topic, microbial spoilage just prior to bottling is near impossible and candidates didn’t grasp this. Some candidates clearly felt that big wineries/facilities need to have a more rigorous bottling hygiene/cleaning regime than small wineries/facilities.

Q3. Describe a comprehensive QA and QC system a large winery should implement for the management of dry goods.

Key understanding:

- *A full written procedure is put in place so all departments at the winery understand the process to ensure dry goods are received, stored and utilised to enable a wine to be produced, bottled and stored in the correct and safe manner, being aware that wine is a food product and consumer safety is paramount.*
- *Dry goods must also match any brand specifications. A QA/QC system must include flow diagrams, HACCP and be able to pass an annual supplier audit and with adequate records for an annual external audit like British Retail Consortium (BRC) or other audits. Storage conditions should also be considered, to avoid damage or contamination.*

This answer required a definition of QA/QC. It also required a definition of dry goods, which should include all packaging goods (bottle, cask or any other vessel, closure, cartons, dividers and labels) plus any products used in the winemaking e.g. yeast, enzyme, tartaric acid, fining agents and oak barrels.

Not many candidates tackled this question, but good answers showed clear practical knowledge and offered good examples. Some candidates wasted too much time discussing procurement procedures or managing supply of goods which are distinct from QA/QC systems. Weak answers just did not cover sufficient ground.

Q4. What are the key factors to consider in drawing up a technical specification for:

a white dessert wine bottled at source with 150 g/l of residual sugar; and an entry-level red wine imported in bulk with 4g/l of residual sugar.

Key understanding

- *Candidates should show understanding that the key objective of technical specifications is to ensure wines arrive with the importer in a good quality and chemically stable condition for sale to the consumer (white) or for final treatment prior to bottling (red).*
- *Should also highlight the potential spoilage which may arise in a very sweet wine and in a dry red wine. Should also understand what a typical specification should cover. There will be differences between a bottled specification and a bulk one.*
- *What are the implications and risks for those specific wine styles (high level of residual sugar etc.) and likely price points? Note that many purchasers may not even consider a technical specification necessary for smaller volumes of a (likely) premium*

sweet wine bottled at source. Highlight what treatments are required for a wine bottled at source and then shipped in bottle and a wine shipped in bulk and prepared for bottling at destination.

This was not a popular question but should have been straightforward as it was quite simply about product specifications – most of which are applicable to both types of wines in the question and then consideration of the different products and risks. Weak answers again talked about quality control in general or offered a list of wine treatments rather than considering what needs to be in a specification and why. Good answers included factual and practical detail.

e) Theory Exam Paper Four – Simon Thorpe MW

Q1. Why do a growing number of large retailers prefer to focus on own and exclusive labels over third-party brands? Is this good for the wine category?

This question covers a subject for which candidates are likely to have been well-prepared, however it is the application of that knowledge which makes the difference between a pass or a fail. The arguments as to why retailers are focusing more on own and exclusive labels were covered well and candidates who had good data concerning growth, margins etc. did well in the first half of the answer. The examiners did find a lot of inaccuracies or errors on statistics though – some of which betrayed a lack of understanding of the market. Better to not quote a fact if you cannot be sure it is correct.

The second half of the question is more subjective and requires a structured and well-argued opinion. Perhaps a SWOT is one way of doing this? No right or wrong answer – but strong analysis led to good scores, lack of insight and too much generalisation led to lower scores.

Q2. How do wine consumers in mainland China decide what wine to buy and what are the implications of their choices for producers and distributors?

Candidates without a good working knowledge of the Chinese market would have been well advised to avoid this question. Besides that, and the ever-present need to be able to present a cogent and plausible argument, this is fundamentally a question with two parts. The first is fact based – around consumer behaviour. The second requires more analysis and possibly brings in comparisons with producer and distributor choices in other more established markets.

Examiners were looking for the ability to be able to demonstrate a thoughtful assessment of the situation and not to generalise. Just knowing the facts is not enough. The pass rate on this question was low, and it was attempted by a small percentage of candidates. Given this is a fundamentally important market for the wine industry, and we are testing knowledge on a global basis, perhaps this performance points to the need for future candidates to focus more on an international aspect to their learning.

Q3. Consider the growth in demand for vegan, organic and sustainable wines. What can and should the wine industry be doing in response?

A "hot" topic and a very popular question which candidates who considered their approach carefully answered well. This is a broad question which requires careful thought on how to structure the answer. The examiners thought a PESTLE approach could be good, and this ought to facilitate comparisons of different markets, different elements in the supply chain, different consumer priorities and different approaches by the wine industry.

Candidates definitely need to remember that this is a Paper 4 question, so although some mention may be made of vineyard and winery practices, this should be dealt with in the context of the commercial implications. For example, if it is accepted that the growth of demand for organic wines will continue, what analysis needs to be undertaken before a winery should consider converting to organic production? What is the premium achieved for organic wines and in which markets is this premium most pronounced?

Good answers demonstrated an understanding of commerical scenarios in a number of markets internationally and were able to analyse the pros and cons well. Those less good answers lacked analysis, spoke solely to facts and tended to concentrate on one market. This question brings in many elements: It showed the need for global understanding and examples, the need to structure an answer carefully, not to flood the answer with too much content and therefore lose clarity and above all it needed analysis.

Q4. Can social media drive brand loyalty in the wine category?

Another "hot" topic – or at least a well-trodden path. The angle here is to concentrate on the concept of brand loyalty – it's no good to simply list a series of successful social media campaigns. The question alludes to wine brands building a community, encouraging their customers to become more engaged with their business and to continue to do this.

Many candidates attempted this question, which is understandable. There were some very good answers demonstrating that candidates could show great and differentiated examples of the commercial success of social media engagement. However, there were many "identikit" answers – somewhat safe answers without evidence of analytical thinking which did not achieve high marks. It's important to remember that a few good, relevant examples from different markets and a clarity of analytical thought go a long way.

Q5. How can the fortified wine category evolve to address current consumer trends?

A strong response to this question on a generally well-understood subject would not have needed a complicated structure. There is a need to quote some data in terms of market growth or decline, and then a need to work out whether to approach the question from the consumer angle or the production angle.

Either approach can be successful, but it seems picking out consumer trends and applying them to existing or possible future product and proposition developments was the optimal approach. Definitely a need to tick a few boxes around increase experiential/tourism opportunity, lighter drinking (cocktails), heritage/authenticity engagement. Miss too many of these out and it becomes difficult to pass. A good number of candidates did cover a lot of the content.

Interestingly, there were some answers which showed a patronising or slightly flippant approach – assumptions around old fashioned businesses unable to adapt themselves to the "modern" consumer. Such generalisations do not indicate that both sides of an issue have been considered.

Q6. Outline the key changes in the fine wine investment market over the past decade. How do you see this developing over the next 10 years?

In order to be successful in this question it was important to have concentrated on the "investment" market and not talked about fine wine in general. Therefore it was essential to be able to quote statistics (size of market in 2009 and 2019) and areas of the global market (both from a production and destination perspective) which have changed most radically in the past decade.

Some essays failed to address the second half of the question at all. Some things to consider would have been whether Bordeaux remains a key element for investment, will Burgundy retain its importance, what other regions might become investment opportunities? Will anything particularly drive investors and if so why?

There were some excellent answers here, candidates with good, in-depth knowledge who addressed the question correctly. Where some candidates fell short was mostly when claims about the future direction of the market were unsubstantiated with any evidence or analysis, or if the candidate simply could not demonstrate enough knowledge.

f) Theory Exam Paper Five - Susan McCraith MW

General Comments

These comments build on those found in previous Examiners' Reports, so do consult those as they are also relevant to this year's exam.

We tend to find that in Paper 5 a lot of marks sit in the 55-64 'Below Threshold' bracket. This is because there are not necessarily right or wrong answers to these questions, it's all about arguing your point authoritatively and insightfully. Opinions need to be based on evidence and attributed and whilst you should come down on one side or the other if you are asked to, it is often appropriate to show the complexity and nuance of certain issues. In other words, we are looking for sophisticated thinking at MW level. You need to demonstrate your skills of analysis, evaluation and argument. Some candidates may have written a comprehensive answer, but not quite passed either because they were more descriptive than critically analytical, or took a simplistic approach, or didn't demonstrate enough experience. The examiner is not just looking for what you know, but also the quality and maturity of your thinking and communication skills.

We have noticed that introductions are getting better year by year. Getting the introduction right is critical. Set out your definitions and what you're going to write about and then stick to this structure. Use signposts to indicate to the reader where they are in the essay e.g. 'turning to the subject of…'. This requires good essay planning which only comes with a lot of practice.

Remember to answer the question set, not leap on the opportunity to deliver a pre-prepared essay, for example on climate change. Give a balanced perspective, show that you have considered all sides and give original examples from around the world.

On a more practical level, you need to write two consistently good essays, there's no point writing one very good one if the average is pulled down by a second, less comprehensive answer. Managing your time in the exam here is key. Leave time to proof-read your work.

Q1. What is the greatest threat currently facing the wine industry and how should this be addressed?

In addressing this question, awareness of various potential threats needed to be shown (climate change is an obvious contender and not to mention it was a serious omission) and a justification given for selecting which one was the greatest. Many candidates understood this and successful passes provided depth with good examples and understanding. They also referenced other threat areas which communicated a good perspective. Definitions needed to be clear. The question specified 'the wine industry' so the entire supply chain needed to be considered. Weak candidates were too narrow and failed to provide enough grasp of the key range and nature of challenges. To concentrate just on one aspect e.g. viticulture led to an incomplete answer. However, there does need to be sufficient detail on the identified key threat. Many people selected climate change here and the temptation to revert to pre-prepared material sometimes meant the focus was not retained on the actual question.

A broad, balanced perspective was required and a recognition that some threats can also be opportunities. Some answers spent too much time on identifying and describing the threat and did not leave enough to talk about solutions, which was an important part of the essay. Arguments needed to be backed up by evidence and examples and not rely solely on opinion or sweeping statements.

Q2. Does a changing climate place greater emphasis on terroir or on choice of grape variety?

Candidates needed to demonstrate the understanding that the climate is changing, not just warming, giving rise to both predictable and unpredictable consequences. This was an interesting question to think about due to the interplay between terroir and grape variety. Answers that fell short of the mark generally did so for failing to answer the question posed i.e. there was too much temptation to describe rather than analyse.

The question required consideration of both aspects but the candidate needed to present a direct answer and strong argument in conclusion, justifying their choice via an analysis of the full range of challenges and opportunities.

Some answers lacked specific examples and were too theoretical, lacking depth and detailed analysis.

Q3. Does wine have a significant role to play in a healthy lifestyle?

This should have been quite a straightforward question but examiners' comments are peppered with comments such as 'superficial', 'lacking in detail', 'rambling'. There were several unsubstantiated, sweeping statements that did not reflect a considered, balanced perspective. A nuanced approach was needed, considering a wide range of different groups and circumstances.

Defining a healthy lifestyle was very important. Answers needed to show knowledge of medical reports and research both for and against wine consumption in a global perspective. A discussion of what constitutes moderation was needed. Wine should have been considered within the context of all alcoholic drinks. There were some unsubstantiated, wild claims. Good answers showed nuance and tackled the conflicting perspectives in a clear and well-structured format.

Q4. How responsible is the wine industry?

This question elicited relatively few responses. Many answers lacked depth or concentrated on one main topic. A definition of 'responsible' was key as well as a consideration of all aspects of the supply chain. Good answers looked at the main ways the industry could be challenged on its responsibility and put wine into context with other similar industries. Social, ethical and environmental issues should have been covered with original examples from all over the world to show a global perspective.

Q5. What makes wine authentic?

This proved a thought-provoking question and the best answers were very interesting to read. There was a necessity to have a broad perspective but also strong views and to back up one's position strongly with key arguments, critical analysis and a rigorous approach. Too few answers took this robust approach and were overly descriptive or lacking in depth. This question led some people to waffle whimsically, while others had too narrow a focus.

Defining 'authentic' was very important. Some people focused solely on natural wines but the question was much broader than this. Good answers encompassed all elements of the supply chain and had fun discussing the different aspects.

Good answers considered why authenticity matters and to whom: authentic to a sense of place, or a style, or a mode of production, to someone's vision, or to a brand identity? You could have looked at whether single vineyards are considered more authentic than regional blends. Whether natural, organic, biodynamic wines are more authentic. Is using clay jars or amphorae more authentic, or just traditional? What about fake wines and counterfeits? How much does the consumer care about authenticity and does that depend on the purchase price? There was much ground to be covered here.